LESSON 3.8

Solve Problems with Unit Rates

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve real-world problems by using unit rates to compare options.

WHY THIS MATTERS You're the new supply manager for the arcade! You need to order tokens, prizes, and snacks from different suppliers. Each supplier offers a different package deal. Your job is to find the unit rate for each deal and figure out which supplier gives you the best value for every item you need to order. Fun Supply Co. sells 500 tokens for $60. Arcade Depot sells 350 tokens for $38.50. Which supplier gives the better value for tokens?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Unit Rate *Spanish: Tasa unitaria*	A rate for just 1 of something, like cost for 1 item.	$2.50 per pound means $2.50 for every 1 pound
Better Buy *Spanish: Mejor compra*	The choice that costs less for each item.	$0.10 each vs $0.15 each — the $0.10 option is the better buy
Comparison *Spanish: Comparación*	Looking at two or more amounts to see how they are related.	$3 per game < $4 per game
Per Unit *Spanish: Por unidad*	For each single item.	$0.50 per token means each token costs $0.50
Rate *Spanish: Tasa*	A ratio comparing two amounts with different units, like miles per hour.	120 miles in 3 hours is a rate (miles and hours are different units)
Proportion *Spanish: Proporción*	A math sentence saying two ratios are equal. It helps find a missing number.	5/3 = x/12 → x = 20

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Fun Supply Co. sells 500 tokens for $60. I want the cost for 1 token.

1. I divide the price by the tokens: $60 ÷ 500 = $0.12.
2. So this deal is $0.12 per token. That is the unit rate.
3. Now I can compare it fairly to a deal with a different number of tokens.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Arcade Depot sells 350 tokens for $38.50. What do we divide?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I find the better buy by comparing the _____ for each option. The option with the _____ unit rate is the better buy because _____.

Word bank Unit Rate | Better Buy | Comparison | Per Unit | Rate | Proportion | unit rate | lower | divide | cost | per | compare

WATCH OUT!

A common mistake

A common mistake in Solve Problems with Unit Rates is comparing the total prices instead of the cost per item

LESSON 3.8

Solve Problems with Unit Rates

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve real-world problems by using unit rates to compare options.

WHY THIS MATTERS You're the new supply manager for the arcade! You need to order tokens, prizes, and snacks from different suppliers. Each supplier offers a different package deal. Your job is to find the unit rate for each deal and figure out which supplier gives you the best value for every item you need to order. Fun Supply Co. sells 500 tokens for $60. Arcade Depot sells 350 tokens for $38.50. Which supplier gives the better value for tokens?

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Fun Supply Co. sells 500 tokens for $60. I want the cost for 1 token.

1. I divide the price by the tokens: $60 ÷ 500 = $0.12.
2. So this deal is $0.12 per token. That is the unit rate.
3. Now I can compare it fairly to a deal with a different number of tokens.

1 **Circle the letter of the best answer. Show how you know.**

Which is the better buy: 6 bouncy balls for $4.50 or 10 bouncy balls for $8.00?

(A) 10 for $8.00 — $0.80 each
(B) They cost the same per ball
(C) Not enough information
(D) 6 for $4.50 — $0.75 each

STEPS TO TRY

1. You can't compare $4.50 and $8.00 straight across — the packs hold different numbers of balls.
2. Find each pack's price for ONE ball (cost ÷ number of balls), then compare the two unit prices.
3. Start with the first deal: $4.50 ÷ 6. Then compute $8.00 ÷ 10 and see which per-ball price is lower.

SHOW YOUR WORK

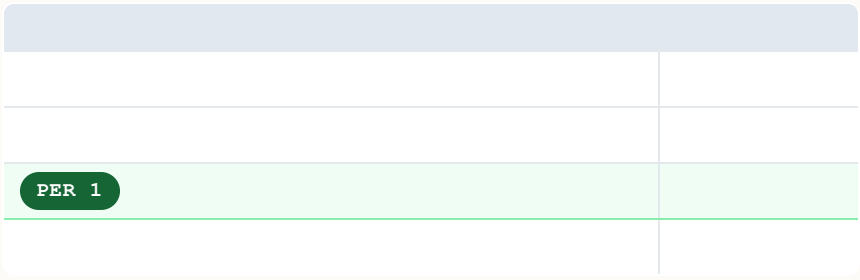
2 **Circle the letter of the best answer. Show how you know.**

A faucet drips 15 ounces of water in 5 minutes. What is the unit rate?

(A) 0.33 ounces per minute
(B) 3 ounces per minute
(C) 5 ounces per minute
(D) 15 ounces per minute

STEPS TO TRY

1. The unit rate here is ounces per ONE minute — 15 ounces spread across 5 minutes.
2. Divide the ounces by the minutes so the time becomes 1 minute.
3. Set up 15 ÷ 5 ounces per minute.



1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

3 **Circle the letter of the best answer. Show how you know.**

A printer prints 42 pages in 6 minutes. How many pages does it print in 15 minutes at the same rate?

(A) 70 pages
(B) 90 pages
(C) 105 pages
(D) 120 pages

STEPS TO TRY

1. This takes two steps: first find pages per ONE minute, then scale up to 15 minutes.
2. Divide to get the unit rate, then multiply that rate by the new time.
3. Step 1: 42 ÷ 6 = 7 pages each minute. Step 2: multiply that rate by 15 minutes.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Store A sells 5 notebooks for $8. Store B sells 3 notebooks for $5.25. Which has the lower price per notebook?

(A) Store B ($1.75 each)

(B) Store A ($1.60 each)

(C) Same price

(D) Cannot determine

STEPS TO TRY

1. Each store sells a different number of notebooks, so compare the price of ONE notebook at each store.
2. Divide cost by quantity for each store, then pick the smaller per-notebook price.
3. Store A: $8 ÷ 5 notebooks. Then compute Store B: $5.25 ÷ 3, and compare the two prices.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do you use unit rates to determine the better buy?

SENTENCE STARTER I find the better buy by comparing the ___ for each option. The option with the ___ unit rate is the better buy because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help